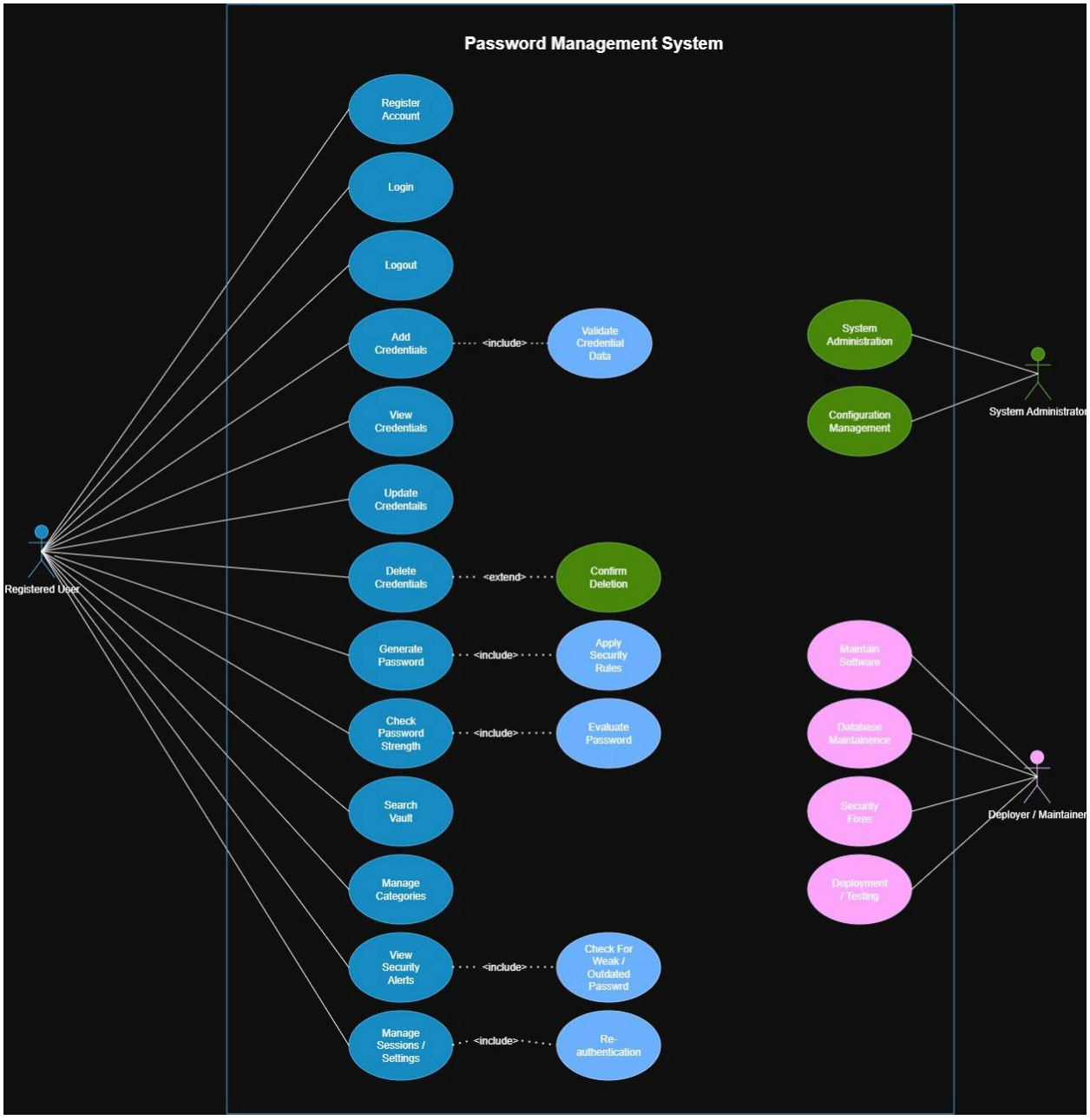
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USE CASE DAIGRAM



2. Roles of Actors

Actor	Role
Registered User	Primary user of the Password Management System. Registers, logs in, manages credentials, generates/checks passwords, searches the vault, manages categories, views alerts, and manages session/settings.
System Administrator	Privileged system operator responsible for limited system administration and configuration. Administrative access does not automatically provide plaintext access to users' vault passwords.
Developer/Maintainer	Technical team member responsible for software development, deployment, database maintenance, testing, and security fixes.

These roles are directly described in the SRS.

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3. Preconditions, Postconditions and Functions of Each Use Case

UC-01: Register Account

Actor: Registered User

Function:
Allows a new user to create an account by providing the required account information.

Precondition:

- User does not already have an account.
- User has access to the registration page.
- Required registration information is available.

Postcondition:

- A new user account is created if the information is valid.
- The authentication password is stored as a secure password hash.
- Duplicate account identifiers are rejected.

The SRS requires validation and secure hashing of authentication passwords.

UC-02: Login

Actor: Registered User

Function:
Authenticates the user and provides access to protected vault functions.

Precondition:

- User has a registered account.
- User provides login credentials.

Postcondition:

- Valid credentials create an authenticated session.
- Invalid credentials result in access being denied.
- The system returns a generic authentication failure message for invalid credentials.

UC-03: Logout

Actor: Registered User

Function:
Terminates the user's authenticated session.

Precondition:

- User is currently authenticated.

Postcondition:

- The active session is invalidated.
- User is returned to an unauthenticated state.

UC-04: Add Credential

Actor: Registered User

Function:
Allows the user to add a website/application credential to the password vault.

Precondition:

- User is authenticated.
- Required credential fields are provided.

Postcondition:

- Credential is validated and stored in protected form.
- Credential is associated with the authenticated user.

Credential information can include service/account name, username, password, category, URL and notes.

UC-05: View Credential

Actor: Registered User

Function:
Allows the user to view their saved credential records.

Precondition:

- User is authenticated.
- Requested credential belongs to the authenticated user.

Postcondition:

- Authorized credential information is displayed.
- Password remains masked by default.

The system must prevent access to another user's vault records.

UC-06: Update Credential

Actor: Registered User

Function:
Allows the user to modify an existing credential.

Precondition:

- User is authenticated.
- Credential exists.
- Credential belongs to the authenticated user.

Postcondition:

- Credential is validated.
- Updated credential information is saved in protected form.

UC-07: Delete Credential

Actor: Registered User

Function:
Allows the user to remove a credential from the vault.

Precondition:

- User is authenticated.
- Credential belongs to the user.
- User confirms deletion.

Postcondition:

- Selected credential is deleted.
- Unrelated records remain unaffected.

The SRS specifically requires **explicit confirmation before credential deletion**.

UC-08: Generate Password

Actor: Registered User

Function:
Generates a strong password using selected length and character-set options.

Precondition:

- User has selected password-generation options.
- At least one character set is selected.

Postcondition:

- A password is generated using a cryptographically secure random source.
- Generated password is displayed to the user.
- It is **not automatically saved**.
- It is saved only if the user explicitly chooses to add it to a credential.

UC-09: Check Password Strength

Actor: Registered User

Function:
Evaluates a password according to configured password-quality rules.

Precondition:

- User provides a password for testing.
- Strength rules are configured.

Postcondition:

- System displays a strength classification such as Weak, Medium, Strong or Very Strong.
- Feedback is provided when the password does not meet policy.
- Tested plaintext password is not stored or logged.

UC-10: Search Vault

Actor: Registered User

Function:
Searches the user's password vault using service/account name or username.

Precondition:

- User is authenticated.
- User has credentials stored in the vault.

Postcondition:

- Matching credentials are displayed.
- Only records belonging to the authenticated user are returned.
- Passwords remain masked by default.

UC-11: Manage Categories

Actor: Registered User

Function:
Allows users to organize credentials using categories.

Precondition:

- User is authenticated.
- User has access to their vault.

Postcondition:

- Credential can be assigned to a category.
- Categories can be created, renamed, deleted or listed.
- Credentials can be filtered by category.

The SRS requires category assignment and category-based filtering.

UC-12: View Security Alerts

Actor: Registered User

Function:
Displays alerts related to weak or outdated passwords.

Precondition:

- User is authenticated.
- Security evaluation has identified relevant credential issues.

Postcondition:

- Security alerts are displayed.
- Password values are not unnecessarily exposed.
- User can mark an alert as reviewed.

UC-13: Manage Session / Settings

Actor: Registered User

Function:
Manages session/security behavior and settings.

Precondition:

- User is authenticated for protected settings.
- User has permission to change the relevant settings.

Postcondition:

- Valid settings are updated.
- Inactive sessions can expire or lock.
- Re-authentication is required after session expiry or vault lock.

The SRS requires authorization checks, session timeout/locking, and re-authentication.